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DRUG EVALUATION

Pharmacokinetic/pharmacodynamic profile of reslizumab in asthma

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ABSTRACT
Introduction: Allergic asthma is a Th2-driven inflammatory process characterized by the infiltration of eosinophils into the airways. IL-5 is a key trigger for eosinophil expansion and release from the bone marrow and plays a crucial role in the entire life span of eosinophils from differentiation to maturation and survival. IL-5 can be considered among the most obvious targets to selectively inhibit eosinophilic airway inflammation.
Areas covered: The preclinical and clinical development of reslizumab, a humanized mAb against IL-5 that has been developed to block IL-5 bioactivity and reduce biologically available IL-5, are described with a particular focus on its pharmacodynamics (PK)/pharmacokinetic (PD) profile.
Expert opinion: Although pivotal trials have documented that reslizumab can be recommended as add-on therapy for the treatment of patients with severe eosinophilic asthma, there is still important information that is lacking and some PK and PD properties of reslizumab are still not fully understood.

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Asthma; eosinophils; humanized monoclonal antibody; interleukin-5; reslizumab

1. Overview of the market

Allergic asthma is a type 2 helper T (Th2)-driven inflammatory process that is characterized by the infiltration of eosinophils into the airways [1]. A high number of eosinophils are found not only in bronchial biopsies but also in sputum and blood of asthmatic patients, especially if they suffer from frequent exacerbations and/or fixed airflow limitation [2]. Eosinophils are recruited to the lungs and airways by cytokines released by activated Th2 cells in combination with a range of chemokines, most notably those of the eotaxin family. They are well accepted as the main effector and tissue-damaging cells during late and chronic stages [3]. In fact, eosinophils contribute to airway wall remodeling and subepithelial membrane thickening via the release of transforming growth factor-β [1], angiogenesis, and bronchial hyperresponsiveness [3]. Consequently, eosinophilic airway inflammation is a key 'treatable trait' in patients with asthma [4].

Interleukin (IL)-5, which is produced by activated Th2 lymphocytes, and in smaller amounts by mast cells, natural killer and natural killer T cells, and eosinophils themselves [5], is a key trigger for eosinophil expansion and release from the bone marrow and plays a crucial role in the entire life span of eosinophils from differentiation to maturation and survival [3]. IL-5 works via the IL-5 receptor (IL-5R), which is composed of two subunits: the α subunit (IL-5Rα) and the β subunit (IL-5Rβ). IL-5Rα is found exclusively on eosinophils and basophils and is responsible for binding IL-5. Hence, IL-5 and its receptor can be considered among the most obvious targets to selectively inhibit eosinophilic airway inflammation [5].

There are currently three monoclonal antibody (mAb) therapies that target the IL-5 or IL-5R pathway in clinical use or undergoing phase III clinical trials in humans [6]. Mepolizumab and reslizumab recognize free IL-5, while benralizumab is a directed at the IL-5Rα (Table 1).

This article focuses on pharmacokinetic (PK)/pharmacodynamic (PD) profile of reslizumab.

2. Characteristics of reslizumab

Initially known as SCH55700 and then as CTx55700 and CEP-38072, reslizumab is a humanized mAb against IL-5 that has been developed to block IL-5 bioactivity and reduce biologically available IL-5, making it clinically beneficial in blocking pathologic inflammatory processes (Box 1).

2.1. Chemistry

Reslizumab, which is a glycoprotein with an approximate a molecular weight of 146 kDa, is based on JES1-39D10, a rat IgG_{2a} antibody with a K_d of 53 pM against human IL-5, and incorporates antigen recognition sites for human IL-5 into consensus regions of a human IgG_4 framework using complementarity-determining region-grafting technology [7].

2.2. Pharmacodynamics

Reslizumab binds specifically to circulating IL-5 and inhibits IL-5-induced cell proliferation [7]. It binds to the small region *ERRR* (glutamic acid, arginine, arginine, and arginine) corresponding to amino acids 89–92 on IL-5, which is critical for binding to IL-5Rα with subsequent inhibition of IL-5 bioactivity [8]

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Box 1. Drug summary.

Drug name	Reslizumab
Phase	Approved for use in the USA, Canada, EU, Norway, Liechtenstein, and Iceland
Indication	Add-on therapy in adult patients with severe eosinophilic asthma inadequately controlled despite high-dose ICSs plus another medicinal product
Mechanism of action	Humanized mAb that binds specifically to IL-5 and inhibits IL-5-induced cell proliferation
Route of administration	Intravenous, subcutaneous (route under investigation, not approved yet)
Molecular formula	Not available
Pivotal studies	3082 and 3083 [15], 3081 [24], 3084 [26]

In vitro studies showed that reslizumab, which has high affinity binding (K_d of 20 pM/l) to IL-5, inhibited the binding of human IL-5 to Ba/F3 cells, which stably express human IL-5 receptor α-subunit, and blocked IL-5-mediated proliferation of human erythroleukemic TF-1 cells [7].

Several *in vivo* experimental models were then used to characterize the reslizumab profile [7]. In allergic mice, reslizumab (0.1–10 mg/kg, via intramuscular [IM] or intraperitoneal [IP] route) not only inhibited the influx of eosinophils into the lungs for a long time (up to 8 weeks after administration of 10 mg/kg and for up to 4 weeks after administration of 2 mg/kg), but also demonstrated an anti-inflammatory activity that was additive to that of oral prednisolone. In sensitized, ovalbumin-challenged guinea pigs, reslizumab at lower doses of (0.03 and 0.3 mg/kg IP) was inactive against the airway hyperreactivity induced by antigen challenge although it significantly inhibited the pulmonary eosinophilia, but at higher doses (up to 30 mg/kg IP), reslizumab inhibited both the pulmonary eosinophilia and airway hyperresponsiveness and, at 30 mg/kg IP, it inhibited allergic, but not histamine-induced bronchoconstriction. Reslizumab (0.1–1 mg/kg IP) also reduced in a dose-dependent manner the pulmonary eosinophilia and neutrophilia in the bronchoalveolar lavage (BAL) fluid caused by tracheal injection of human IL-5 in guinea pigs. In ovalbumin-sensitized rabbits, reslizumab (intravenous [IV] 5 mg/kg) administered 2 h before ovalbumin challenge significantly reduced cutaneous eosinophilia but did not affect the total number of cells and neutrophils in the skin. In cynomolgus monkeys challenged with *Ascaris suum* extract, a single dose of reslizumab (0.3 mg/kg IV) reduced the eosinophilia in the BAL fluid caused by antigen challenge for up to six months, with 75% inhibition of eosinophil accumulation.

After administration of reslizumab at 0.3 mg/kg in patients with severe persistent asthma, a short-lived decrease in circulating blood eosinophil counts was observed varying from a mean reduction of 52.5% at 48 h to 18.9% at Day 30, compared with baseline [9]. The decrease obtained with reslizumab at 1.0 mg/kg was more pronounced and remained significant up to Day 30. There was a tendency toward reduction of eosinophils in induced sputum after administration of reslizumab at 1 mg/kg.

2.3. Pharmacokinetics and metabolism

In mice, serum levels of reslizumab were 69,157 ng/ml 2 h after 10 mg/kg IP and fell to 6476 ng/ml by 8 weeks after treatment [7]. When reslizumab was given at 2 mg/kg IP, its serum levels were 19,077 ng/ml at 2 h and fell to 2068 ng/ml by 4 weeks after treatment.

An initial human study evaluated the PK trend of rising single doses of reslizumab (0.03 mg/kg, 0.1 mg/kg, 0.3 mg/kg, or 1.0 mg/kg) administered IV in patients with severe persistent asthma [9]. The plasma concentrations of reslizumab were dose proportional. Its mean systemic exposure (area under curve [AUC]) increased in a 1–2.2–5.9–23 proportion as the dose increased in a 1-3.3–10–33.3 proportion. The mean maximal concentration obtained 6.9 h after reslizumab dosing at 1.0 mg/kg was 30.3 µg/ml. Mean concentrations were 0.87 µg/ml on Day 90 and 0.43 µg/ml on Day 120 after 1.0 mg/kg, suggesting that this could represent a threshold value of biological activity. The elimination half-life ranged between 24.5 and 30.1 days.

A large summary of the PK of reslizumab is available in the product monograph [10] and a recent review (Table 2) [11]. The PK properties of IV reslizumab, which were characterized in volunteers (n = 130), patients with asthma (n = 438) or nasal polyps (n = 236), were similar across these subjects, with variability in peak and overall exposure between individuals of about 20–30%. Peak serum concentrations usually occurred at the end of the infusion and generally declined from this peak in a biphasic manner. The drug accumulates approximately 1.5- to 1.9-fold in the serum following multiple doses. A population PK model has recently documented that the overall steady-state exposures of reslizumab were comparable across a greater than fourfold range of weight (33.9 to <63 kg, 63 to <73 kg, 73 to <85.5 kg, and 85.5–156 kg) [12]. Reslizumab has a volume of distribution of approximately 5 L, suggesting minimal distribution to the extravascular tissues. Similar to other mAbs, it is likely that reslizumab is degraded by enzymatic proteolysis into small peptides and

Table 1. Humanized anti-interleukin (IL)-5 antibodies.

Drug	Mechanism of action	Effects
Reslizumab	It binds to the small region *ERRR* (glutamic acid, arginine, arginine, and arginine) corresponding to amino acids 89–92 on IL-5, which is critical for binding to IL-5Rα with subsequent inhibition of IL-5 bioactivity.	Reduction in eosinophil production, activity, and survival.
Mepolizumab	It specifically binds to the α-chain of IL-5 and disrupts its interaction with the α-subunit of the IL-5R, which is present on the eosinophils and their early precursors.	Dramatic and sustained decrease in blood eosinophilia (including $CCR3^+$ cells), decreased eosinophil activation, and increased circulating levels of IL-5.
Benralizumab	It binds to a specific epitope within the extracellular domain on IL-5Rα, which is in close proximity to the IL-5-binding site and thus inhibits IL-5R signaling, independent of ligand.	Depletion of eosinophils and basophils because of apoptotic destruction and decreased production of these cells because of the blockade of IL-5R signaling, independent of ligand.

Table 2. Summary of pharmacokinetic parameters of reslizumab.

Dose	Parameter					
	$C_{max,ss(0–4wk)}$	$AUC_{ss(0–4wk)}$	$C_{av,ss(0–4wk)}$	CL	V_d	$t_{1/2}$
3.0 mg/kg IV	86.27–105.33 μg/ml	27.2–33.1 mg·h/ml	40.4–49.3 μg/ml	≈7 ml/h	≈5 L	≈24 days

Data from [11] and [12].

$AUC_{ss(0–4wk)}$: area under the reslizumab serum concentration versus time curve; $C_{av,ss(0–4wk)}$: average reslizumab serum concentration; CL: systemic clearance; $C_{max,ss(0–4wk)}$: maximum reslizumab serum concentration; $t_{1/2}$: elimination half-life; V_d: volume of distribution.

amino acids. As reslizumab binds to a soluble target, linear non-target-mediated clearance is expected. Its clearance is approximately 7 ml/h and is expected to be linear and non-target mediated. Reslizumab has a half-life of about 24 days.

No clinical studies have been conducted to assess the impact of hepatic impairment or renal impairment on the PK of reslizumab. However, since kidneys and liver are not involved in the disposition of mAbs, hepatic and renal impairments are not likely to alter PK enough to justify dosage adjustment. In such cases, according to the Food and Drug Administration (FDA) guidance, a study to confirm that prediction may be helpful, but is not necessary [13].

No formal clinical drug interaction studies have been performed with reslizumab. However, *in vitro* data indicate that IL-5 and reslizumab are unlikely to affect CYP1A2, 2B6, or 3A4 enzyme activity [11]. In effect, population PK analyses indicate that the concomitant use of leukotriene antagonists or corticosteroids had no significant effect on its PK.

2.4. Pharmacogenetics

No specific studies were performed to assess the impact of race or ethnicity on the PK of reslizumab, but a population analysis demonstrated no significant differences in the PK of reslizumab with regard to race (Caucasian, Black, and Asian) [11].

2.5. Clinical efficacy

2.5.1. Early trials

In an initial small study that enrolled patients with symptomatic severe persistent asthma despite using high-dose inhaled corticosteroids (ICSs) or oral corticosteroids, reslizumab at a single IV dose raging between 0.03 and 1.0 mg/kg reduced the number of circulating blood eosinophils in subjects with severe persistent asthma [9]. In patients who received 1.0 mg/kg, the response was more profound and the effect was sustained for 30 days. In addition, it induced a limited short-lived improvement in baseline forced expiratory volume in 1 s (FEV_1).

A 15-week, phase II, randomized, double-blind, placebo-controlled clinical study evaluated the effect of reslizumab (3 mg/kg IV every 4 weeks per four doses) in patients with asthma and elevated sputum eosinophil counts (at least 3%) that was poorly controlled with ICSs [14]. Reslizumab reduced sputum and blood eosinophils and improved airway function particularly in patients with nasal polyps. Modest improvement in asthma control was also observed with reslizumab, but the difference between the treatment groups did not reach statistical significance. There was also a trend toward improvement in the exacerbation rates in the reslizumab group compared to placebo: 8% versus 19% of the patients had at least one exacerbation, respectively.

2.5.2. Pivotal trials

The phase III BREATH program evaluated reslizumab treatment in four separate trials involving more than 1700 patients with elevated eosinophils, whose symptoms were inadequately controlled with ICSs with or without another controller medication.

Two large, multicenter, double-blind, randomized, placebo-controlled, 52-week phase III trials enrolled 953 patients who had at least one asthma exacerbation requiring systemic corticosteroid use over the past 12 months, had to have at least one blood eosinophil count of 400/μl, and were receiving at least medium-dose ICS with or without a second controller (studies 3082 and 3083) [15]. Maintenance oral corticosteroid (up to 10 mg per day prednisone equivalent) was allowed. The patients received either 13 doses of placebo or reslizumab 3 mg/kg administered IV once every 4 weeks. Reslizumab significantly lowered the annual rate of clinical asthma exacerbations by 54% (50% and 59% in studies 1 and 2, respectively) in comparison with placebo. It also prolonged the time to first exacerbation with a probability of not having an exacerbation of 61% and 73% in studies 1 and 2 compared to 44% and 52% in the placebo group, respectively. A reduction in asthma exacerbations resulting in hospitalization or an emergency room visit was observed with reslizumab 3 mg/kg that was not statistically significant (34% and 31% in studies 1 and 2, respectively). Larger reductions in exacerbation rates were noted in patients with repeated exacerbations in the period of 12 months before the trial. Moreover, in both trials, reslizumab significantly decreased blood eosinophil numbers, significantly improved patient-reported outcomes (Asthma Control Questionnaire-7 [ACQ-7] and Asthma Quality of Life Questionnaire scores), and increased FEV_1 values.

A secondary analysis of these data examined the efficacy of reslizumab as add-on therapy in patients receiving asthma treatment per Step 4 and Step 5 of the Global Initiative for Asthma (GINA) guidelines at baseline was performed [16]. The results of this analysis showed that this mAb reduces exacerbation rates and improves lung function and patient-reported outcomes in patients requiring either GINA Step 4 or Step 5 therapy. It was also documented that reslizumab produces larger reductions in asthma exacerbations and larger improvements in lung function in patients with exacerbation-prone, late-onset asthma (an age cutoff of ≥40 years) with elevated blood eosinophil levels (≥400 cells/μl) and inadequately controlled symptoms, which was larger than that observed for the overall study population or patients with age of onset <40 years [17]. Additionally, pooled results from these two studies showed that reslizumab was effective at reducing the frequency of asthma exacerbations and in improving lung function, asthma control, and quality of life in patients ≥65 years old [18]. The observed improvements in this subgroup were numerically larger than for younger adults. Notably, patients who demonstrated an early FEV_1 response (defined as ≥100 ml) or early ACQ response to reslizumab before 16 weeks had a greater improvement in clinical asthma exacerbation rate relative to placebo after 52 weeks than patients who did not respond according to either criterion by week 16 (59–76% versus 26–31% reduction relative to placebo) [19]. Intriguingly, high-risk asthma patients with low baseline lung function (FEV_1 percent predicted <50%) derived significant therapeutic benefit from reslizumab as evidenced by the reduction in exacerbations requiring systemic corticosteroids and improvement in FEV_1 [20]. It was also noted that patients with inadequately controlled asthma, elevated blood eosinophils, and aspirin sensitivity with or without chronic rhinosinusitis with nasal polyps received significant therapeutic benefit, providing further support for the use of reslizumab in this clinical setting [21].

Interestingly, in one of the two studies (study 3082), a 42% reduction in exacerbation rates was seen with reslizumab in the omalizumab-eligible patient population (defined as total IgE of 30–700 IU/ml, body weight <150 kg, and perennial allergen sensitization), indicative of the clinically meaningful benefit of reslizumab in patients eligible for treatment with either biologic treatment [22]. In any case, in an indirect treatment comparison using a network meta-analysis to compare exacerbation rate ratios of reslizumab to omalizumab and mepolizumab, reslizumab resulted in lower exacerbation rates compared to omalizumab and mepolizumab [23].

A third study was a 16-week, randomized, placebo-controlled trial (study 3081). There was no prior asthma exacerbation requirement for this study. Maintenance oral corticosteroid was not allowed [24]. Three hundred and eleven patients, aged 12–75 years with asthma inadequately controlled by at least using medium-dose ICS and with a blood eosinophil count ≥400 cells/μl, received either four doses of placebo or reslizumab 0.3 or 3 mg/kg administered IV once every 4 weeks. After 16 weeks of therapy, patients who received reslizumab had clinically significant improvements in pulmonary function (increases in FEV_1 of 0.286 and 0.242 l were reported in the reslizumab 3.0 and 0.3 mg/kg groups, respectively, while the placebo group demonstrated an FEV_1 change of 0.126 l), quality of life, and self-reported asthma control compared to placebo as well as a decrease in the use of inhaled rescue medications. The response was observed as early as 4 weeks, and a significant decrease in the blood eosinophil count was appreciated. Even though there was a significant response with the 0.3 mg/kg dose, the effect was considered superior with the 3.0 mg/kg dosing scheme.

Eligible patients who had participated in one of these three phase III reslizumab studies (studies 3081, 3082, and 3083) were given the option to enter a long-term, open-label extension study in which they received reslizumab 3 mg/kg IV once every 4 weeks for up to 2 years [25]. The study enrolled 1051 patients. Continuous exposure, including during the placebo-controlled studies, was ≥12 months for 740 patients and ≥24 months for 249 patients. Lung function was more favorable at the start of the extension in patients who had received reslizumab than in those who had received placebo in the prior trials and this was maintained throughout the extension study. However, reslizumab-naive patients experienced FEV_1 increases by week 4 of active treatment and improvements were maintained throughout the extension study period in all patients. Blood eosinophil counts appeared to be returning to baseline after reslizumab discontinuation.

Another 16-week, randomized, double-blind, placebo-controlled, multicenter phase III trial explored the use of reslizumab (3 mg/kg IV every 4 weeks per four doses) in 492 uncontrolled asthmatics with a wide range of blood eosinophil levels (study 3084) [26]. Approximately 80% of patients had an eosinophil count <400 cells/μl. There were not significant changes in FEV_1 in the overall study population and in the subgroup of patients with a blood eosinophil count of <400 cells/μl. Linear regression analysis failed to demonstrate a significant relationship between baseline blood eosinophils and change in FEV_1. However, in the subgroup of patients with a blood eosinophil threshold of ≥400 cells/μl, with respect to placebo, reslizumab produced a significant mean FEV_1 increase (an average FEV_1 0.076 l greater than placebo). The pattern of improvements in other efficacy parameters (quality of life, self-reported asthma control compared to placebo, and use of inhaled rescue medications) at 16 weeks was consistent with that observed for FEV_1. A marked decrease in blood eosinophils was observed after the first dose of reslizumab compared with placebo and was maintained during the 16 weeks.

Data from BREATH program were pooled from subgroups of patients with early eosinophil assessment [27]. A substantial decrease in blood eosinophil levels was detected as early as 2–3 days after the first dose of reslizumab, although blood eosinophil levels remained low but detectable throughout 16 weeks of reslizumab doses.

The recent documentation that four doses of 3.0 mg/kg IV reslizumab were superior to fixed-dose of 100 mg subcutaneous (SC) mepolizumab in attenuating airway eosinophilia in 13 patients with prednisone-dependent (5–30 mg per day of prednisone or its equivalent) eosinophilic asthma (sputum eosinophilia >3% and blood eosinophils >300/μl) with associated improvement in asthma control and FEV_1 is extremely interesting because it suggests that more drug may have reached the airway when administered by the IV route [28].

2.5.3. Ongoing trials

A phase II trial that aims to explore if reslizumab is able to improve the clinical condition of patients with severe asthma who failed to respond to omalizumab because of lack of efficacy (symptoms-Asthma Control Tesr (ACT) score <20, exacerbations) or adverse effects (ClinicalTrials.gov identifier: NCT03074942) is currently recruiting participants.

Also an exploratory phase IV study (Distribution of Eosinophils in Asthma After Reslizumab, [DEAR]) with the primary objectives to (1) establish that PET/CT of the lung can reliably distinguish healthy, non-asthmatic volunteers from patients with asthma and an eosinophilic phenotype and (2) examine the utility of PET/CT for demonstrating that reslizumab produces a reduction in lung inflammation in patients with severe asthma and an eosinophilic phenotype (ClinicalTrials.gov identifier: NCT02937168) is recruiting patients.

The primary objective of a 52-week, double-blind, placebo-controlled, parallel-group, efficacy and safety study (ClinicalTrials.gov identifier: NCT02452190) is to determine the effect of SC reslizumab 110 mg administered every 4 weeks on clinical asthma exacerbations in adults and adolescents with asthma and elevated blood eosinophils who are inadequately controlled on standard-of-care asthma therapy. The study is ongoing, but not recruiting participants.

The primary objective of a phase III, 24-week, double-blind, placebo-controlled, parallel-group study (ClinicalTrials.gov identifier: NCT02501629), which is ongoing, but not recruiting participants, is to assess the ability of SC reslizumab 110 mg administered every 4 weeks to produce a corticosteroid-sparing effect in patients aged ≥12 years with oral corticosteroid-dependent asthma and elevated blood eosinophils, without loss of asthma control.

2.6. Safety, tolerability, and toxicity

No significant adverse effects were observed in CD-1 mice and cynomolgus monkeys administered reslizumab intravenously up to dose levels of 25 mg/kg QM for 6 months [10]. No carcinogenicity was observed in a 6-month bioassay in which reslizumab was administered intravenously once every 2 weeks for 26 consecutive weeks (14 total doses) to Tg. rasH2 mice at doses up to 516 mg/kg/dose. In a fertility study, administration of reslizumab to parental CD-1 mice at doses up to 50 mg/kg (approximately six times the maximum recommended human dose on an AUC basis) had no effects on male or female mating or fertility.

In general, reslizumab appears to be safe and is well tolerated in the clinical studies [10]. There is no evidence of increased incidence of adverse events (AEs) that might be related to long-term exposure in patients treated for up to 36 months [24]. A recent systematic review and meta-analysis assessed the proportion of individuals who discontinued the treatment due to AE using data from five randomized clinical trials (RCTs) representing a cohort of 1365 participants (713 in the reslizumab group and 652 in the placebo group) [29]. The result suggests that reslizumab and placebo were similar in terms of the incidence of withdrawn due to AEs.

The most commonly experienced AEs included oropharyngeal pain (3%), an increase in levels of the enzyme creatine phosphokinase in the blood (a measure of possible damage to muscles) (2%), and myalgia (1%). Malignant neoplasms occurred in a small number of patients (0.6% compared with 0.3% of those in the placebo group) [30,31]. These responses were diverse and not associated with any particular tissue type.

The most important risk associated with the use of reslizumab is anaphylaxis, which was reported in 0.3% of the patients in the clinical studies [30] and was always antidrug antibody negative. These events were observed during or within 20 min after completion of the reslizumab infusion and reported as early as the second dose of this mAb. Manifestations included dyspnea, decreased oxygen saturation, wheezing, vomiting, and skin and mucosal involvement, including urticaria [32]. Anaphylaxis can be life threatening [10]. Patients should be observed for an appropriate period of time after the IV infusion of reslizumab. More postmarketing data will be needed to draw any conclusions. However, anaphylaxis is the subject of a boxed warning in the labeling for reslizumab.

2.7. Regulatory affairs and approvals

On March 2016, the US FDA approved reslizumab for use with other asthma medicines for the maintenance treatment of severe asthma in patients aged 18 years and older with an eosinophilic phenotype [33]. Reslizumab granted approval for patients who have a history of severe asthma attacks (exacerbations) despite receiving their current asthma medicines with a boxed warning. In fact, since reslizumab can cause serious side effects including allergic (hypersensitivity) reactions that can be life threatening, it should be administered in a health-care setting by a health-care professional who is prepared to manage anaphylaxis.

On July 2016, Health Canada granted approval of reslizumab IV for the add-on maintenance treatment of adult patients with severe eosinophilic asthma who are inadequately controlled with medium-to-high-dose ICSs and an additional asthma controller(s) (e.g. long acting β_2 agonist [LABA]) and have a blood eosinophil count of ≥400 cells/μl at initiation of the treatment [34].

On August 2016, the European Commission granted marketing authorization for reslizumab IV to be used as an add-on therapy in adult patients with severe eosinophilic asthma inadequately controlled despite high-dose ICSs plus another medicinal product in the 28 countries of the European Union (EU) in addition to Norway, Liechtenstein, and Iceland [35].

The British National Institute for Health and Care Excellence (NICE) [36] has recommended reslizumab add-on therapy as an option for the treatment of severe eosinophilic asthma that is inadequately controlled in adults despite maintenance therapy with high-dose ICSs plus another drug, only if (1) the blood eosinophil count has been recorded as 400 cells/μl or more; (2) the person has had three or more asthma exacerbations in the past 12 months; and (3) the company provides reslizumab with the discount agreed in the patient access scheme. At 12 months, reslizumab must be stopped if the asthma has not responded adequately or continued if the asthma has responded adequately. Response must be assessed each year.

3. Conclusion

Reslizumab is a humanized mAb capable to attach to IL-5, which stimulates the growth and activity of eosinophils, and prevents binding of IL-5 to its receptor. Consequently, it reduces the number of eosinophils in the blood and lungs. Pivotal trials have documented that reslizumab improves pulmonary function and reduces asthma exacerbations in patients with elevated sputum eosinophils or blood eosinophil counts ≥400 cells/μl who cannot be adequately controlled with high doses of ICSs and another medicine used for the prevention of asthma. Overall it is well tolerated. Safety and efficacy have not been established in patients younger than 18 years of age.

Based on efficacy and safety data from phase II/III clinical trials, reslizumab had been approved for use as an add-on maintenance treatment of severe asthma with an eosinophilic phenotype in adults who have a history of exacerbations despite receiving their current asthma medicines in the USA, Canada, and EU in addition to Norway, Liechtenstein, and Iceland. However, it carries a boxed warning for anaphylaxis; therefore, patients will require drug administration and subsequent observation by a health-care professional who is able to treat an anaphylactic reaction.

Ongoing trials are exploring the effects of SC reslizumab in severe asthma.

4. Expert opinion

Although pivotal trials have documented that reslizumab can be recommended as add-on therapy for the treatment of patients with severe eosinophilic asthma, there is still important information that is lacking and some PK and PD properties of reslizumab are still not fully understood.

Unfortunately, a full population PK analysis that could support optimal drug utilization by quantifying typical disposition characteristics and sources of variability (such as between-subject variability) within study populations has not been published yet, although some very partial information has been included in a monograph [10] and a review article [11].

In any case, there are no PK data on airway bioavailability of reslizumab in patients with asthma. Furthermore, although it has been reported that population PK analyses indicate that the concomitant use of leukotriene antagonists or corticosteroids had no significant effect on its PK [10], we believe that the current information about relevant drug–reslizumab interaction is not exhaustive and more formal studies are needed in this field. Regrettably, the clinical studies of reslizumab do not provide sufficient data to determine whether the drug poses a risk during pregnancy. Moreover, it is not known whether reslizumab is present in human milk, and the effects of reslizumab on breastfed infants and on milk production are not known. Still, no clinical studies have been conducted to assess the effects of renal or hepatic impairment on the PK of reslizumab.

The approved dosing of reslizumab is based on patient weight at 3 mg/kg once every four weeks via IV infusion over 20–50 min. This should be considered an advantage not only because IV administration of biotech drugs offers the advantage of circumventing presystemic degradation, thereby achieving the highest concentration in the biological system. Actually, the drawbacks of SC and IM injections include a potentially decreased bioavailability that is secondary to variables such as local blood flow, injection trauma, protein degradation at the site of injection, and limitations of uptake into the systemic circulation related to effective capillary pore size and diffusion [37]. Nonetheless, some patients and clinicians may prefer the more convenient SC dosing [38]. Additionally, patients with obesity may require larger doses of reslizumab, which may impact the cost of therapy. Therefore, there are some studies in which fixed-dose SC preparation will be used to assess this alternative dosing route.

The availability of both routes of administration will certainly benefit, especially to ensure patient compliance with the treatment. However, it will be greatly valued if the PK profiles produced by the two routes of administration will be compared in the same patients. In this way, it will be possible not only to determine the PD equivalence, if any, of the doses used in the two routes, but it may also be found out if there are particular conditions favoring the choice of a route of administration rather than another, beyond the preference of the patient.

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